

# VulMorph-Fed: Cross-Project Software Vulnerability Detection via Federated Vulnerability Morphology Learning

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## Abstract

Deep learning-based software vulnerability detection faces two coupled barriers: models fail to generalise across projects because they overfit project-specific lexical patterns, and organisations cannot pool the proprietary code that would diversify training data. We propose VulMorph-Fed, a federated learning framework built on the observation that vulnerability manifestations vary lexically across projects while their structural characteristics (morphologies) are largely shared. VulMorph-Fed combines three components: Vulnerability-Critical Sub-graph Abstraction (VCSA), which maps code tokens onto a small, fully specified taxonomy of abstract operation types and learns to down-weight vulnerability-irrelevant edges; Morphology-Conditioned Federated Prototype Aggregation (MCFPA), which shares CWE-conditioned prototypes instead of model parameters under a provable per-round  $\varepsilon$ -differential-privacy guarantee with explicit composition accounting; and Morphology-Guided Message Passing (MGMP), which injects the aggregated global prototypes into local graph neural network training. We evaluate on project-level cross-project splits of four public C/C++ corpora (Devign, BigVul, DiverseVul, PrimeVul) against baselines organised by privacy class and matched in data, budget, backbone and input representation, reporting mean and standard deviation over repeated runs with statistical tests. We compare against DP-FedAvg at an identical per-round budget, report both F1 and AUC, and communicate  $\sim 11$  KB per client per round. We also report, candidly, where the design loses: non-private parameter averaging on the same abstracted graphs remains the stronger aggregator when no privacy constraint is imposed.

The complete implementation and scripts regenerating every reported number are publicly released.

*Keywords:* Program Representation, Static Analysis, Abstract Syntax Trees, Software Vulnerability Detection, Federated Learning, Graph Neural Networks, Differential Privacy

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## 1. Introduction

Software vulnerabilities are a critical and growing threat to modern computing systems. The National Vulnerability Database (NVD) reported over 25,000 new Common Vulnerabilities and Exposures (CVEs) in 2024 alone, highlighting the urgent need for automated and scalable vulnerability detection mechanisms. While Deep Learning-based Vulnerability Detection (DLVD) methods have demonstrated significant promise, they continue to suffer from two fundamental barriers that impede their adoption in real-world scenarios: the generalisation gap and the data silo problem.

First, state-of-the-art DLVD models exhibit a catastrophic *generalisation gap* when applied across different software projects [1]. For instance, a model trained on the source code of FFmpeg typically fails to reliably detect vulnerabilities in OpenSSL, even when both projects manifest the identical Common Weakness Enumeration (CWE) type. Replication studies consistently report performance drops of 50% to 90% in cross-project evaluation settings compared to within-project evaluations [2, 3]. This degradation occurs because existing models tend to overfit to project-specific lexical patterns (e.g., API names, variable naming conventions) rather than learning the underlying structural semantics of the vulnerability.

Second, the *data silo problem* prevents the aggregation of sufficient training data. Real-world vulnerability data is highly sensitive and routinely proprietary. Organisations such as financial institutions, cloud service providers, and defence contractors cannot legally or practically share their raw source code with external entities due to intellectual property concerns and privacy regulations.

25 Consequently, a centralised DLVD model trained on one organisation’s codebase cannot leverage the knowledge embedded in another organisation’s repositories, severely restricting the model’s exposure to diverse vulnerability manifestations.

Federated Learning (FL) has emerged as a promising paradigm to address the data silo problem by enabling collaborative model training without centralising raw data [4, 5]. However, naively applying standard FL techniques, such as FedAvg, to vulnerability detection does not resolve the generalisation gap [6]. Weight averaging across highly heterogeneous, non-IID (Independent and Identically Distributed) codebases fails to transfer structural vulnerability knowledge effectively, leading to suboptimal cross-project detection performance [7, 8].

The root cause of both barriers stems from the representation level: existing approaches primarily model code at the lexical or token level rather than at the structural or semantic level [9, 10]. A buffer overflow in FFmpeg and a buffer overflow in OpenSSL are structurally identical within their Code Property Graphs (CPGs)—both typically involve a loop incrementing an index, an array access using that index, and a missing bounds check—yet they utilise completely different function names and API calls.

To overcome these challenges, we propose **VulMorph-Fed** (Vulnerability Morphology-Guided Federated Learning), a novel federated learning framework designed to achieve robust cross-project vulnerability detection while preserving data privacy. VulMorph-Fed operates on the principle that vulnerability patterns are structurally invariant across projects but lexically heterogeneous. By extracting and federating *vulnerability morphologies* rather than raw parameters or lexical features, VulMorph-Fed successfully bridges the generalisation gap without requiring centralised code sharing.

The primary contributions of this paper are:

- We introduce the **Vulnerability-Critical Subgraph Abstraction (VCSA)**, which combines a fully specified, deterministic mapping of code tokens onto a small taxonomy of abstract semantic operation types (given ex-

55       haustively in Table 1) with a learned, trainable edge-weighting module  
that down-weights vulnerability-irrelevant structure.

- We propose **Morphology-Conditioned Federated Prototype Aggregation (MCFPA)**, an FL aggregation mechanism that shares CWE-conditioned semantic prototypes instead of model parameters. We prove a  
60       per-round  $\epsilon$ -differential-privacy guarantee for the released prototype bank (Proposition 1) with explicit norm clipping, per-class noise calibration, and sequential-composition accounting over rounds.
- We design **Morphology-Guided Message Passing (MGMP)** to inject global prototype signals into local Graph Neural Network (GNN) training,  
65       allowing clients to benefit from structural knowledge of CWE types they have never observed locally.
- We evaluate on project-level cross-project splits of four public corpora with multi-seed repetition, statistical testing, and a baseline suite organised by *privacy class*: methods with a formal  $\epsilon$ -DP guarantee (DP-FedAvg  
70       at our exact per-round budget) are compared as peers, while centralised training and parameter-sharing FedAvg — including FedAvg on our own abstracted representation — serve as non-private references. This design isolates the representation from the federation mechanism and prices privacy explicitly. We release the complete implementation and the scripts  
75       that regenerate every reported number.<sup>1</sup>

The remainder of this paper is structured as follows. Section 2 reviews related work in deep learning for vulnerability detection, cross-project prediction, and federated learning. Section 3 details the proposed VulMorph-Fed framework, including its label model, privacy analysis, and inference protocol. Section  
80       4 presents the experimental setup and results. Section 5 discusses implications and limitations, and Section 6 concludes.

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<sup>1</sup><https://github.com/vinhqdang/vulmorph-fed>

## 2. Background and Related Work

### 2.1. Deep Learning for Software Vulnerability Detection

The application of Deep Learning (DL) to Software Vulnerability Detection (SVD) has evolved significantly from early sequence-based models to modern graph-based and transformer-based architectures. Early works, such as VulDeePecker [9] and SySeVR [11], treated code as sequences of tokens, employing recurrent neural networks (RNNs) like BiLSTMs to classify code gadgets. While these sequence-based models demonstrated the feasibility of DL for SVD, they struggled to capture the complex, non-linear structural dependencies inherent in source code.

The introduction of the Code Property Graph (CPG) [12], which unifies the Abstract Syntax Tree (AST), Control Flow Graph (CFG), and Data Flow Graph (DFG), marked a paradigm shift. Graph Neural Networks (GNNs) became the architecture of choice for SVD because they naturally align with the topological structure of CPGs. Devign [10] pioneered this approach by applying Gated Graph Neural Networks (GGNNs) over composite graphs to detect vulnerabilities in large C/C++ projects. Subsequent GNN-based models have focused on improving graph representation and interpretability. For instance, DeepWukong [13] operates on inter-procedural program slices, while IVDetect [14] provides fine-grained interpretations using GNNExplainer. More recent advancements include Vul-LMGNNs [15], which fuses Large Language Models (LLMs) with GNNs, and HAGNN [16], which utilises heterogeneous attention mechanisms.

Simultaneously, Transformer-based models and LLMs have been adapted for SVD. LineVul [17] leverages CodeBERT for line-level vulnerability prediction, and LLMxCPG [18] uses CPGs to guide LLM attention. Despite these advances, recent studies [19, 20] indicate that LLMs often operate at a shallow structural level, highlighting the continued necessity for deep structural abstractions. Furthermore, rigorous replication studies [1, 2, 3] have consistently exposed a severe generalisation gap: SOTA models perform well in within-project settings but their accuracy drops precipitously when evaluated on unseen projects (cross-

project).

## 2.2. Cross-Project Vulnerability and Defect Prediction

Cross-Project Vulnerability Detection (CPVD) and Cross-Project Defect  
115 Prediction (CPDP) aim to train models on a source project and deploy them on  
a target project. Traditional CPDP often relies on software metrics and transfer  
learning [7]. In the context of DLVD, several works have employed Domain  
Adaptation (DA) techniques. CD-VulD [21] and VulGDA [22] use deep domain  
adaptation on graph embeddings to achieve cross-domain discovery. CPVD [23]  
120 combines Graph Attention Networks (GATs) with adversarial domain adaptation,  
representing the current SOTA for fully supervised cross-project SVD. CSVD-TF [24]  
fuses expert and semantic metrics using TrAdaBoost. However, a critical limitation  
of these DA-based approaches is that they require centralised access to raw source  
code from both the source and target domains, violating  
125 privacy constraints in many real-world scenarios.

## 2.3. Federated Learning in Software Engineering

Federated Learning (FL) [4] allows multiple clients to jointly train a global  
model without sharing their local data. Its application to Software Engineering  
(FL-SE) is a burgeoning area of research, motivated by the proprietary nature  
130 of commercial codebases [25]. Recent empirical studies [26, 8] confirm the feasibility  
of FL for tasks like code clone detection and vulnerability detection.

However, adapting FL to SVD presents unique challenges, primarily due  
to severe non-IID data distributions (heterogeneity in CWE types and coding  
styles). FedGraphNN [6] demonstrated that standard federated GNNs often  
135 underperform centralised models on non-IID graph splits. Existing FL-SVD  
solutions typically rely on NLP models (e.g., CodeBERT) or naive GNN aggregations  
[27, 8], which inherit the generalisation gap of their centralized counterparts.  
To address graph heterogeneity, approaches like FedDense [28] decouple structural  
and feature learning. VulMorph-Fed builds upon these concepts but  
140 specifically targets the structural invariance of vulnerabilities by federating semantic  
prototypes rather than raw model parameters.

#### 2.4. Privacy-Preserving Mechanisms

Ensuring data privacy in FL often involves Differential Privacy (DP) [5] or Secure Aggregation [29]. DP algorithms, such as DP-SGD, add calibrated noise (e.g., Laplace or Gaussian) to model updates. The magnitude of this noise is proportional to the sensitivity of the function being computed. Standard DP-SGD on large GNNs or LLMs requires significant noise due to the high dimensionality of the model parameters and the large vocabulary size of source code tokens, severely degrading utility. VulMorph-Fed mitigates this by applying DP at the level of semantic prototypes. By utilizing a compact, abstracted morphological space instead of the full token vocabulary, the sensitivity is drastically reduced, enabling a much more favorable privacy-utility trade-off.

### 3. Methodology

To address the cross-project generalisation gap and the data-silo problem simultaneously, we propose *VulMorph-Fed*. The framework replaces standard parameter averaging with a privacy-preserving federated aggregation of semantic structural prototypes. It consists of three components: (1) Vulnerability-Critical Subgraph Abstraction (VCSA), which maps functions onto project-invariant structural morphologies; (2) Morphology-Conditioned Federated Prototype Aggregation (MCFPA), which aggregates structural knowledge under differential privacy; and (3) Morphology-Guided Message Passing (MGMP), which injects the aggregated global knowledge into local graph learning. Figure 1 illustrates the end-to-end abstraction on two lexically different but structurally similar vulnerable functions.

#### 3.1. Graph Construction

Let  $\mathcal{G}_k = \{(G_i^{(k)}, y_i^{(k)}, c_i^{(k)})\}_{i=1}^{N_k}$  denote the local dataset of client  $k \in \{1, \dots, K\}$ , where  $y_i \in \{0, 1\}$  is the binary vulnerability label and  $c_i$  the CWE annotation of vulnerable functions (Section 3.3).

Full Code Property Graph (CPG) extraction requires a heavyweight static-  
 170 analysis engine such as Joern [12], which is impractical to deploy at every fed-  
 erated edge client and complicates exact reproduction. Our implementation  
 instead parses every function with `tree-sitter` — a production-grade, error-  
 tolerant C/C++ parser — and builds the graph  $G = (\mathcal{V}, \mathcal{E})$  directly from the  
 abstract syntax tree:

- 175 • one node per *named AST node*, in pre-order, capped at  $|\mathcal{V}| \leq 200$ ;
- *syntax edges*  $\mathcal{E}_{AST}$ : parent-child links in both directions;
- *order edges*  $\mathcal{E}_{SIB}$ : consecutive named siblings (source-order sequencing,  
 the AST analogue of Devign’s NCS edges [10]);
- *data-dependence proxy edges*  $\mathcal{E}_{DD}$ : def-use chains linking successive oc-  
 180 currences of the same identifier leaf.

The edge set is  $\mathcal{E} = \mathcal{E}_{AST} \cup \mathcal{E}_{SIB} \cup \mathcal{E}_{DD}$ ; on our corpora this yields on av-  
 erage  $\sim 130$  nodes and  $\sim 350$  edges per function. Functions that fail to parse  
 (a small minority) fall back to a token-level sequence graph and are marked  
 as such. These graphs carry real syntactic structure, though not the interpro-  
 185 cedural control- and data-flow precision of a compiler-grade CPG; Section 5  
 discusses this trade-off, and graph construction is isolated behind a single inter-  
 face so Joern-derived CPGs can be substituted without touching the rest of the  
 framework.

### 3.2. Vulnerability-Critical Subgraph Abstraction (VCSA)

190 Raw code graphs carry thousands of project-specific API and identifier to-  
 kens, causing GNNs to overfit local lexical distributions. VCSA replaces the  
 lexical node labels with a small taxonomy of abstract operation types and learns  
 to down-weight vulnerability-irrelevant edges.

#### 3.2.1. Morphological taxonomy

195 We define a fixed semantic taxonomy  $\mathcal{T}$  of  $|\mathcal{T}| = 8$  operation types, listed  
 exhaustively in Table 1 together with the reserved fallback type UNKNOWN. A



Table 1: The complete  $|\mathcal{T}| = 8$  morphological taxonomy (plus the reserved `UNKNOWN` fallback). A node takes one of  $|\mathcal{T}| + 1 = 9$  morphological *categories*; each category is mapped to a learned embedding (dimension 64 in our configuration), summed with the node’s grammar-kind embedding.

| Abstract type               | Covers                                                                                                                                                                                                                      |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>MEMORY_ACCESS</code>  | allocation/free/copy/set and string APIs ( <code>malloc</code> , <code>free</code> , <code>memcpy</code> , <code>strcpy</code> , ...)                                                                                       |
| <code>ARRAY_INDEX</code>    | array subscript operations <code>a[i]</code>                                                                                                                                                                                |
| <code>PTR_DEREF</code>      | pointer dereference and member access ( <code>*p</code> , <code>p-&gt;f</code> , unary <code>&amp;</code> )                                                                                                                 |
| <code>CONTROL_BRANCH</code> | <code>if/else/switch</code> , loops, <code>goto/break/continue/return</code> , ternary                                                                                                                                      |
| <code>ARITH_OP</code>       | arithmetic and bitwise operators ( <code>+</code> <code>-</code> <code>*</code> <code>/</code> <code>%</code> <code>&amp;</code> <code> </code> <code>^</code> <code>~</code> <code>&lt;&lt;</code> <code>&gt;&gt;</code> ) |
| <code>COMPARISON</code>     | relational and logical operators ( <code>==</code> <code>!=</code> <code>&lt;</code> <code>&gt;</code> <code>&lt;=</code> <code>&gt;=</code> <code>&amp;&amp;</code> <code>  </code> <code>!</code> )                       |
| <code>CALL_SITE</code>      | any other function/macro call (user-defined and unresolved calls included)                                                                                                                                                  |
| <code>ASSIGN</code>         | plain and compound assignments ( <code>=</code> , <code>+=</code> , ...)                                                                                                                                                    |
| <code>UNKNOWN</code>        | all remaining tokens (identifiers, literals, punctuation, declarations)                                                                                                                                                     |

node therefore takes one of  $|\mathcal{T}| + 1 = 9$  morphological categories — the `UNKNOWN` type has its own feature slot. For the taxonomy-size sensitivity analysis (RQ2b) we additionally define strictly nested refinements with  $|\mathcal{T}| = 16$  and  $|\mathcal{T}| = 32$  (e.g., `MEMORY_ACCESS` splits into `MEMORY_ALLOC`, `MEMORY_FREE`, `MEMORY_COPY`, `STRING_OP`); the complete lists ship with the source code.

Each node’s feature vector is the sum of two embeddings, both drawn from *project-invariant* vocabularies: (i) its morphological type in  $\mathcal{T} \cup \{\text{UNKNOWN}\}$ , and (ii) its *grammar kind* — the tree-sitter node category (`call_expression`, `subscript_expression`, `if_statement`, ...), a fixed set of 364 categories determined entirely by the C grammar and therefore identical for every project and client by construction. No identifier, API name, literal or any other project-specific token ever enters the network.

### 3.2.2. The abstraction function $\phi_{\text{type}}$

The mapping  $\phi_{\text{type}} : \mathcal{V} \rightarrow \mathcal{T} \cup \{\text{UNKNOWN}\}$  is a deterministic rule set evaluated on the AST, in priority order:

1. a `call_expression` is classified by its callee name in two stages: an exact match against fixed libc allow-lists (`malloc`, `memcpy`, `strcpy`, ...), then

a set of *affix rules* that recognise project-wrapped equivalents. The affix stage is essential rather than cosmetic: production C code almost never calls libc allocators directly (FFmpeg uses `av_malloc`, GLib `g_malloc`, the Linux kernel `kmalloc`, OpenSSL `OPENSSL_malloc`), so an exact-match list alone routes nearly every real allocation site to `CALL_SITE`. We measured this: on Devign, exact matching alone assigns `MEMORY_ACCESS` to only 0.23% of AST nodes, rising to 0.72% once affix rules are applied — i.e. two thirds of the memory-safety signal the taxonomy is designed to capture is invisible without them. The rules test for the affixes `*alloc*`, `*free*`, `*memcpy*`, `*memset*`, `*str{cpy,cat,len}*` and are evaluated in an order that resolves `realloc` and `dealloc` before the generic `alloc` test. Only the matched *class* is retained — never the identifier — so project invariance is preserved. Unmatched calls, including all user-defined functions, macros and unresolved calls, map to `CALL_SITE`; no name resolution or build system is required;

2. statement and expression kinds map directly: `subscript_expression` → `ARRAY_INDEX`; `field_expression` and unary pointer expressions → `PTR_DEREF`; `if/switch/loop/jump` statements and ternaries → `CONTROL_BRANCH`; `assignment_expression` and initialising declarators → `ASSIGN`;
3. a `binary_expression` maps by its operator text to `ARITH_OP` or `COMPARISON` (Table 1);
4. three further kind rules complete the set: `cast_expression` → `PTR_DEREF` (a cast reinterprets a pointee type), `update_expression` (`i++`, `i--`) → `ARITH_OP`, and unary `!` → `COMPARISON` with the remaining unary operators → `ARITH_OP`;
5. every remaining node maps to `UNKNOWN`.

An equivalent token-context rule set (documented in the code) covers the rare parse-failure fallback graphs. The rules are implemented once at the finest ( $|\mathcal{T}| = 32$ ) granularity and projected onto the coarser taxonomies, so all three taxonomy sizes share a single rule set.  $\phi_{\text{type}}$  is a discrete label assignment and is

not differentiable; the only learned part of VCSA is the edge-weighting module

below.

On our four evaluation corpora, 25–26% of AST nodes receive a non-UNKNOWN morphological type (the remainder are declarations, identifiers, literals and other syntax carriers, which still contribute their grammar-kind feature). The abstraction does not delete nodes: all nodes are retained with their graph structure, but their features are collapsed from a  $\sim 10,000$ -token lexical vocabulary onto  $|\mathcal{T}| + 1$  morphological categories plus 364 grammar kinds. This many-to-one collapse is what removes project-specific lexical information from everything the model computes and, downstream, from everything a client transmits (Section 3.4.1).

### 3.2.3. Learned edge weighting

VCSA additionally learns a soft edge-importance mask. A two-layer MLP over concatenated endpoint features produces

$$\varepsilon_e = \sigma(\text{MLP}([\mathbf{x}_u \parallel \mathbf{x}_v])) \in [0, 1] \quad \text{for } e = (u, v) \in \mathcal{E}, \quad (1)$$

which reweights messages during propagation (Eq. 7); edges are softly down-weighted rather than hard-pruned, keeping the pipeline trainable end-to-end. Sparsity is encouraged by an  $\ell_1$  penalty  $\mathcal{L}_{\text{sparse}} = \frac{1}{|\mathcal{E}|} \sum_e \varepsilon_e$ . To align morphologies across clients we further use a structural contrastive loss over graph embeddings  $\mathbf{h}_G$ :

$$\mathcal{L}_{SCL} = \frac{1}{|\mathcal{P}^+ \cup \mathcal{P}^-|} \left[ \sum_{(i,j) \in \mathcal{P}^+} \|\mathbf{h}_{G_i} - \mathbf{h}_{G_j}\|_2^2 + \sum_{(i,j) \in \mathcal{P}^-} \max(0, m - \|\mathbf{h}_{G_i} - \mathbf{h}_{G_j}\|_2^2) \right], \quad (2)$$

where  $\mathcal{P}^+$  are same-CWE vulnerable pairs and  $\mathcal{P}^-$  are (vulnerable, benign) pairs within a mini-batch, with margin  $m = 1$ . The total local objective is  $\mathcal{L} = \mathcal{L}_{BCE} + \alpha \mathcal{L}_{SCL} + \gamma \mathcal{L}_{\text{sparse}}$  with  $\alpha = 0.1$ ,  $\gamma = 0.01$ ;  $\mathcal{L}_{BCE}$  is class-weighted binary cross-entropy with  $\text{pos\_weight} = \min(N^-/N^+, 20)$  computed on each client’s local split.

**Project A (FFmpeg-style):**

```
n = av_get_size(ctx);
buf = av_malloc(n);
for (i = 0; i <= n; i++)
    buf[i] = src[i];
```

**Project B (OpenSSL-style):**

```
len = BN_num_bytes(a);
out = OPENSSL_malloc(len);
for (k = 0; k <= len; k++)
    out[k] = in[k];
```

**Morphology sequence emitted by  $\phi_{\text{type}}$  — identical for both fragments:**

```
CONTROL_BRANCH → ASSIGN → ASSIGN → ASSIGN → COMPARISON → ARITH_OP → CALL_SITE →
MEMORY_ACCESS → ASSIGN → ARRAY_INDEX → ARRAY_INDEX
```

Figure 1: Worked example of the VCSA abstraction. Two off-by-one buffer overflows from different projects share no API or identifier token — `av_get_size`/`BN_num_bytes`, `av_malloc`/`OPENSSL_malloc`, `buf`/`out` — yet  $\phi_{\text{type}}$  maps both onto a byte-identical morphology: a size query (`CALL_SITE`), an allocation (`MEMORY_ACCESS`), and a loop whose *inclusive* bound (`COMPARISON` on  $\leq$ ) drives an array write (`ARRAY_INDEX`). The sequence shown is the literal output of the released `classify_token/_classify_ast_node` implementation on these two inputs (breadth-first over the AST, `UNKNOWN` nodes elided), not an idealisation; it is regenerated from the code so that figure and implementation cannot diverge. Note that recognising `av_malloc` and `OPENSSL_malloc` as allocations requires the affix rules of Section 3.2 — an exact libc allow-list maps both to `CALL_SITE` and the shared morphology is lost.

### 3.3. Label Model

Real vulnerability corpora have a many-to-many CVE–function–CWE structure, so we state our label model explicitly:

- 270 • **Detection is strictly binary.** The classification head predicts  $\hat{y} \in [0, 1]$  (vulnerable vs. benign); all reported Precision, Recall, F1 and AUC values are computed from this binary head. The decision threshold is calibrated on held-out training-project data as described in Section 4, never on the test set; AUC is threshold-free. CWE information never enters the detec-  
275 tion head.
- **CWE labels condition only the prototypes.** A vulnerable function annotated with several CWEs contributes to the prototype of its *first-listed* (*primary*) CWE. Benign functions contribute to a dedicated *benign* prototype slot, giving the bank an explicit model of non-vulnerable mor-  
280 phology.

• **CWE vocabulary.** The prototype bank has  $|\mathcal{C}| + 1 = 11$  slots: the nine most frequent CWE types in the training corpus receive dedicated slots, all remaining types share an `OTHER` slot, and the eleventh slot is the benign prototype. This bucketing is computed once per dataset and reported with the released configurations. Datasets without per-function CWE annotations (Devign) degenerate gracefully to a single vulnerable-prototype slot plus the benign slot.

### 3.4. Morphology-Conditioned Federated Prototype Aggregation (MCFPA)

FedAvg-style parameter averaging [4] assumes near-IID clients; vulnerability data is extremely non-IID in its CWE composition. MCFPA instead federates *class-conditioned prototypes* over the  $|\mathcal{C}| + 1$  bank slots of Section 3.3 (CWE buckets plus the benign slot). After local training, client  $k$  computes, for each slot  $c$  with local support  $N_{c,k} > 0$ ,

$$\mathbf{p}_c^{(k)} = \frac{1}{N_{c,k}} \sum_{G \in \mathcal{G}_k: y_G=1, c_G=c} \text{clip}_R\left(f_{\theta_k}(\tilde{G})\right), \quad \text{clip}_R(\mathbf{z}) = \mathbf{z} \cdot \min\left(1, \frac{R}{\|\mathbf{z}\|_1}\right), \quad (3)$$

i.e., the centroid of the  $\ell_1$ -clipped embeddings of its local functions assigned to slot  $c$  (vulnerable functions of the corresponding CWE bucket; all benign functions for the benign slot). Slots with  $N_{c,k} = 0$  are transmitted as zero rows and carry no information.

#### 3.4.1. Differential privacy

Before upload, each non-empty prototype row is perturbed with the Laplace mechanism:

$$\hat{\mathbf{p}}_c^{(k)} = \mathbf{p}_c^{(k)} + \text{Lap}\left(0, \frac{2R}{N_{c,k} \varepsilon}\right)^{\otimes d}. \quad (4)$$

VulMorph-Fed applies two mechanisms, so that the guarantee covers the whole pipeline rather than only the released artefact.

*Mechanism 1: private encoder training..* Local training uses DP-SGD [5]: each per-example gradient is clipped to  $\|g_i\|_2 \leq C$  and the batch sum is perturbed with Gaussian noise of standard deviation  $\sigma C$  before the optimiser step. The

encoder  $\theta_k$  is therefore itself a differentially private function of the client’s data. This matters: without it, everything below would be conditional on an encoder that had seen the raw data unprotected, and changing one record could move every embedding by an unbounded amount.

310 *Mechanism 2: private prototype release..* Prototypes are built from  $\ell_1$ -clipped embeddings (Eq. 3) and perturbed per class (Eq. 4).

**Proposition 1** (Per-round privacy of the prototype release). *Fix a client  $k$  and condition on its (DP-trained) encoder. Let two neighbouring datasets differ in one function. Releasing the noisy bank  $\{\hat{\mathbf{p}}_c^{(k)}\}$  of Eq. (4) satisfies  $2\varepsilon$ -differential*  
 315 *privacy with respect to that record.*

*Proof.* With the encoder fixed, each per-sample embedding lies in the  $\ell_1$  ball of radius  $R$ , so replacing one record changes an affected class mean by at most  $\Delta_1 = 2R/N_{c,k}$  in  $\ell_1$  norm; the Laplace mechanism with scale  $\Delta_1/\varepsilon$  is  $\varepsilon$ -DP for that row [30]. Under the replacement neighbouring relation a record change can  
 320 affect *two* rows — it leaves the slot of its old label and joins the slot of its new one — so parallel composition across slots does not apply and the two affected rows compose sequentially, giving  $2\varepsilon$ . (A restricted neighbouring relation that preserves the label would give  $\varepsilon$ ; we report the weaker, safer constant.)  $\square$

**Theorem 1** (End-to-end budget). *Let  $\varepsilon_{\text{sgd}}$  be the budget consumed by DP-SGD*  
 325 *over all local steps, accounted under Rényi differential privacy for the subsampled Gaussian mechanism [31] at a target  $\delta$ . Then  $T$  rounds of VulMorph-Fed satisfy*

$$\varepsilon_{\text{tot}} = \varepsilon_{\text{sgd}} + 2T\varepsilon \quad (5)$$

*( $\varepsilon_{\text{tot}}, \delta$ )-differential privacy with respect to any single function in a client’s dataset, covering both the transmitted prototypes and the locally trained encoder.*

*Proof.* DP-SGD over the local steps is  $(\varepsilon_{\text{sgd}}, \delta)$ -DP by the Rényi accountant; the  
 330  $T$  prototype releases are  $2T\varepsilon$ -DP by Proposition 1 and sequential composition. The two mechanisms observe the same private dataset, so their budgets add.  $\square$

We use the standard Rényi accountant (the binomial bound of Mironov et al. [31] for the sampled Gaussian, converted to  $(\varepsilon, \delta)$  over a grid of orders) rather than advanced composition, which would overstate  $\varepsilon_{\text{sgd}}$  by more than an order  
 335 of magnitude; our implementation reproduces the reference regime of Abadi et al. [5] to within the expected margin.

*Residual leakage outside the analysis..* Two channels remain, and we state them rather than leaving them implicit. The noise scale  $2R/(N_{c,k}\varepsilon)$  depends on the private count  $N_{c,k}$ , and the *support pattern* of a client’s occupied slots is  
 340 transmitted unnoised, revealing which weakness classes a client possesses at all. Padding the bank to a fixed support and using a count-independent scale would close both at some utility cost; we quantify neither here.

We emphasise why the morphological abstraction matters here: because the encoder consumes only  $|\mathcal{T}|+1$  abstract node types, the embeddings being clipped  
 345 and averaged contain no project-specific lexical information to begin with. DP protects *membership* of individual functions; the abstraction removes *content* (identifiers, API names) categorically rather than statistically.

### 3.4.2. Affinity-weighted aggregation

The server aggregates the noisy banks per CWE bucket. Let  $A_k \subseteq \{1..K\}$   
 350 be the clients with non-zero rows for bucket  $c$ . With  $\bar{\mathbf{p}}_j = \hat{\mathbf{p}}_c^{(j)} / \|\hat{\mathbf{p}}_c^{(j)}\|_2$ , the server computes cosine-affinity weights and the global prototype

$$w_c^{(k)} = \max\left(0, \frac{1}{|A_k| - 1} \sum_{j \in A_k, j \neq k} \bar{\mathbf{p}}_j^\top \bar{\mathbf{p}}_k\right), \quad \mathbf{p}_c^{\text{glob}} = \frac{\sum_{k \in A_k} w_c^{(k)} \hat{\mathbf{p}}_c^{(k)}}{\sum_{k \in A_k} w_c^{(k)}}, \quad (6)$$

falling back to the uniform mean when all affinities vanish, and to the previous round’s value when no client has data for  $c$ . Clients whose local view of CWE  
 355  $c$  agrees with the consensus contribute more; outlier (or heavily noised) views are attenuated. The ablation “w/o MCFPA” replaces Eq. (6) with the uniform mean.

### 3.5. Morphology-Guided Message Passing (MGMP)

Each client integrates the broadcast bank  $\mathcal{P}^{\text{glob}} = \{\mathbf{p}_c^{\text{glob}}\}$  into its GNN. Layer  $l$  first performs edge-weighted local aggregation (GCN-style normalisation 360  $\hat{d}$  with the VCSA mask  $\varepsilon$  as edge weight):

$$\mathbf{h}_v^{\text{loc}} = \sum_{u \in \mathcal{N}(v) \cup \{v\}} \frac{\varepsilon_{uv}}{\sqrt{\hat{d}_u \hat{d}_v}} \mathbf{W}^{(l)} \mathbf{h}_u^{(l)}, \quad (7)$$

then attends over the global prototypes with scaled dot-product attention:

$$\mathbf{m}_v = \sum_{c \in \mathcal{C}} \text{softmax}_c \left( \frac{(\mathbf{W}^{(l)} \mathbf{h}_v^{(l)})^\top \mathbf{p}_c^{\text{glob}}}{\sqrt{d}} \right) \mathbf{p}_c^{\text{glob}}, \quad (8)$$

and fuses the two signals through a per-node morphology gate  $\beta_v = \sigma(\mathbf{w}_\beta^\top [\mathbf{W}^{(l)} \mathbf{h}_v^{(l)} \parallel \mathbf{x}_v^{\text{morph}}])$  and a learnable per-layer balance  $\lambda \in [0, 1]$ :

$$\mathbf{h}_v^{(l+1)} = \text{ReLU}((1 - \lambda) \mathbf{h}_v^{\text{loc}} + \lambda \beta_v \mathbf{m}_v). \quad (9)$$

Nodes whose abstract type is implicated in known vulnerability morphologies 365 receive a higher gate  $\beta_v$  and thus a stronger global signal. Even a client that has never observed CWE  $c$  locally can recognise its structural morphology by attending to  $\mathbf{p}_c^{\text{glob}}$ .

The global bank additionally reaches the decision through two further channels. First, the graph-level readout is  $[\mathbf{h}_G^{\text{mean}} \parallel \mathbf{h}_G^{\text{max}} \parallel \cos(\mathbf{h}_G^{\text{mean}}, \mathbf{p}_c^{\text{glob}})_{c \in \mathcal{C} \cup \{\text{ben}\}}]$ : 370 the classifier directly sees how similar the current function is to every global morphology, including the benign one. Second, a prototype-alignment regulariser

$$\mathcal{L}_{\text{proto}} = \frac{1}{|B|} \sum_{i \in B} \|\mathbf{h}_{G_i} - \mathbf{p}_{s(i)}^{\text{glob}}\|_2^2, \quad (10)$$

where  $s(i)$  is sample  $i$ 's bank slot (its CWE bucket, or the benign slot), pulls each client's embedding space toward the shared global geometry — the mechanism FedProto [32] showed to be essential for prototype-based FL under non-IID 375 clients. The total local objective becomes  $\mathcal{L} = \mathcal{L}_{BCE} + \alpha \mathcal{L}_{SCL} + \gamma \mathcal{L}_{\text{sparse}} + \mu \mathcal{L}_{\text{proto}}$  with  $\mu = 0.5$ .



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**Algorithm 1** VulMorph-Fed

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**Require:**  $K$  clients, rounds  $T$ , per-round budget  $\varepsilon$ , clip radius  $R$

```
1: Server initialises  $\mathcal{P}^{\text{glob},(0)} = \mathbf{0}$ 
2: for  $t = 1, \dots, T$  do
3:   for each client  $k$  in parallel do
4:     Retrieve  $\mathcal{P}^{\text{glob},(t-1)}$ ; abstract graphs via VCSA ( $\phi_{\text{type}}$ , Eq. 1)
5:     Train  $\theta_k$  for  $E$  local epochs with MGMP (Eqs. 7–9) and loss  $\mathcal{L}_{BCE} +$   

 $\alpha\mathcal{L}_{SCL} + \gamma\mathcal{L}_{\text{sparse}} + \mu\mathcal{L}_{\text{proto}}$  (Eq. 10)
6:     Build clipped prototypes  $\mathbf{p}_c^{(k)}$  (Eq. 3); perturb via Eq. (4)
7:     Upload  $\{\hat{\mathbf{p}}_c^{(k)}\}_{c \in \mathcal{C}}$   $\{2|\mathcal{C}|d$  floats per round incl. download $\}$ 
8:   end for
9:   Server aggregates  $\mathcal{P}^{\text{glob},(t)}$  via Eq. (6) and broadcasts
10: end for
Ensure: Prototype bank  $\mathcal{P}^{\text{glob},(T)}$ ; ensemble detector of Eq. (11); spent budget  

 $\varepsilon_{\text{tot}} = T\varepsilon$ 
```

---

### 3.6. Training and Inference Protocol

Algorithm 1 summarises one federation. Note that *model parameters are never aggregated*: each client keeps its own encoder  $\theta_k$ , and the only shared  
380 state is the prototype bank.

**Inference.** The deployed global detector is the uniform probability ensemble of the  $K$  client models, each conditioned on the same final prototype bank:

$$\hat{p}(y \mid G) = \frac{1}{K} \sum_{k=1}^K \sigma\left(f_{\theta_k}(\tilde{G}; \mathcal{P}^{\text{glob}})\right). \quad (11)$$

All headline metrics are computed from Eq. (11) on the held-out cross-project test set; we additionally report the mean and standard deviation of per-client  
385 F1 to expose client-level variance. In a production deployment, Eq. (11) corresponds to clients exposing scoring endpoints, or — when a single model is required — any client model can be used alone, at the cost quantified by the reported per-client variance.

### 3.7. Communication Cost

390 Per round, each client uploads and downloads one  $(|\mathcal{C}| + 1) \times d$  float32 bank:  
2  $\times 11 \times 128 \times 4 \text{ B} \approx 11 \text{ KB}$  per client per round in our configuration, independent  
of model size; total server traffic is  $K$  times that (e.g.,  $\approx 220 \text{ KB}$  per round at  
 $K = 20$ ). Standard FedAvg on the same backbone transmits the full parameter  
vector ( $\sim 1.4 \text{ MB}$  for our deliberately compact GNN; hundreds of MB for code-  
395 LM backbones) per client per round.

## 4. Experimental Evaluation

We design the evaluation to answer five research questions:

- **RQ1 (Cross-Project Generalisation):** How does VulMorph-Fed compare to centralised and federated baselines — including baselines that  
400 share its backbone or its input representation — on held-out projects never seen by any client?
- **RQ1b (Value of Federation):** What do private federated clients add on top of a model trained centrally on public data alone?
- **RQ2 (Ablations):** What do VCSA, MCFPA and MGMP contribute  
405 individually, and how sensitive is the framework to the taxonomy size  $|\mathcal{T}|$ ?
- **RQ3 (Privacy-Utility):** How does detection quality degrade across per-round DP budgets  $\varepsilon$ , accounting for composition over rounds?
- **RQ4 (Scalability):** How do performance and communication cost be-  
410 have as the number of clients  $K$  grows?

### 4.1. Experimental Setup

#### 4.1.1. Datasets

We evaluate on four public C/C++ vulnerability corpora with complementary characteristics:

- 415 1. **Devign** [10]: 21,854 manually labelled functions from FFmpeg and QEMU ( $\approx 46\%$  vulnerable); no per-function CWE annotations.
2. **BigVul** [33]: CVE-linked functions from 234 projects (in our sample), with CWE annotations; realistic class imbalance ( $\approx 6\%$  vulnerable).
3. **DiverseVul** [34]: the most project-diverse corpus (580 projects in our sample, 59 CWE types;  $\approx 5.5\%$  vulnerable).
- 420 4. **PrimeVul** [3]: the strictest recent benchmark, with rigorous de-duplication and label verification.

Devign is loaded from its public HuggingFace mirror, BigVul and DiverseVul from the `bstee615` mirrors, and PrimeVul from the `ASSERT-KTH` mirror (un-  
 425 paired training split); exact dataset revisions are pinned in the released code. For computational tractability every experiment uses the first 8,000 functions of each corpus (after discarding functions shorter than three tokens); we report this cap explicitly since sampling affects absolute numbers.

#### 4.1.2. Cross-project federated split

430 All splits are at *project* granularity: every function of a repository is assigned to exactly one partition. Projects are sorted by sample count and the smallest projects are held out until they total  $\approx 20\%$  of samples — these held-out projects form the test set, and *no function from any test project is ever seen by any client*. The remaining training projects are assigned round-robin (by project)  
 435 to  $K = 4$  clients, so each client holds a disjoint set of repositories, simulating per-organisation codebases. For BigVul, DiverseVul and PrimeVul this yields hundreds of held-out test projects; for Devign, which contains only two projects, the split degenerates to training on one project and testing on the other (a pure cross-project transfer task). Every RQ1 experiment is repeated with five random  
 440 seeds  $\{42, \dots, 46\}$  (three for the secondary analyses). The seed draws a *fresh project-level holdout* as well as the client assignment, model initialisation and DP noise, so the reported standard deviations include split variance — which, in cross-project evaluation, dominates initialisation variance. We report mean

± standard deviation throughout, and per-seed values feed the statistical tests  
445 below.

#### 4.1.3. Baselines

The baseline suite is organised by *privacy class*, and within each class it isolates the two factors that could otherwise confound the comparison — input representation and aggregation mechanism.

450 **Non-private references** (raw code pooled, or parameters exchanged in the clear — vulnerable to gradient-inversion and membership attacks, hence *not* peers of a DP system):

- **Centralised GGNN / GAT (lexical graphs)**: Devign-style [10] and CPVD-style [23] backbones on pooled raw data;
- 455 • **Centralised Transformer (sequence)**: a Transformer encoder over raw token sequences (structure-free family, LineVul-style [17]); a compact model trained from scratch, not a pre-trained code LM — pre-trained LM baselines are left to future work;
- **Centralised GGNN (VCSA graphs)**: the GGNN backbone on our  
460 abstracted representation, isolating representation from protocol;
- **FedAvg + GAT / Transformer (lexical)**: standard federated baselines;
- **FedAvg + GGNN (VCSA graphs)**: FedAvg on the *same* input representation as VulMorph-Fed;
- 465 • **Centralised VulMorph (oracle)**: our full model on pooled data, no federation, no DP.

**Formally private peers** (per-round  $\varepsilon$ -DP, matched to VulMorph-Fed’s budget):

- 470 • **DP-FedAvg + GAT / + GGNN (VCSA graphs)**: the standard route to formal privacy for parameter-sharing FL — each client’s model

update is norm-clipped and perturbed with a Laplace mechanism calibrated to the same per-round  $\varepsilon$  as our prototype release. Both mechanisms spend the same per-round  $\varepsilon$ , but they spend it very differently: a single  $\ell_1$  clipping budget must cover the entire parameter update (0.19–0.67M  
475 coordinates depending on backbone), whereas ours covers an  $(|\mathcal{C}| + 1) \times d = 1,408$ -coordinate bank whose rows are additionally means over  $N_{c,k}$  records. The per-coordinate signal-to-noise ratio therefore differs by orders of magnitude. We report this as the operative mechanism rather than attributing the effect to dimensionality *per se*: for the Laplace mechanism  
480 the per-coordinate noise scale is  $\Delta_1/\varepsilon$  and does not depend on  $d$  directly.

All baselines use the same data, splits, seeds, class-weighted loss, downsampling, threshold calibration, optimiser and training budget as VulMorph-Fed; FedAvg aggregation is sample-size weighted.

#### 4.1.4. Implementation details

485 All models are implemented in PyTorch / PyTorch Geometric; graphs are built with tree-sitter as described in Section 3.1. VulMorph-Fed uses a 2-layer GNN, hidden dimension  $d = 128$ , embedding dimension 64, mean $\oplus$ max readout, dropout 0.3, Adam with learning rate  $10^{-3}$ , batch size 64,  $T = 10$  federated rounds with  $E = 2$  local epochs,  $|\mathcal{C}| = 10$  CWE buckets, taxonomy size  $|\mathcal{T}| =$   
490 8, per-round budget  $\varepsilon = 2.0$  and clip radius  $R = 1$  unless stated otherwise. Two class-imbalance measures apply identically to every system (ours and all baselines): each client’s *training* split caps the benign:vulnerable ratio at 4:1 (test and calibration splits keep true prevalence), and the loss is class-weighted with  $\text{pos\_weight} = \min(N^-/N^+, 20)$ . Inference follows the ensemble protocol  
495 of Eq. (11). The decision threshold of every system is calibrated by maximising F1 on a calibration slice (10% of each client’s training-project samples, held out before downsampling); *the test set never influences the threshold*, and AUC is threshold-free.

#### 4.1.5. Statistical testing

500 For each baseline we pair the per-dataset, per-seed F1 values of VulMorph-Fed with those of the baseline (up to  $4 \times 3 = 12$  pairs) and run a two-sided Wilcoxon signed-rank test with Cliff’s  $\delta$  effect size; we mark a comparison significant at  $p < 0.05$ . With three seeds per dataset the per-dataset tests are underpowered on their own, which is why we test across the pooled dataset–  
505 seed pairs and additionally release all raw per-seed values in the artifact.

#### 4.1.6. Reproducibility

The complete implementation — loaders, models, federated runners, baselines, and the scripts that regenerate every table and figure in this paper from scratch (`experiments/run_all.sh`) — is publicly available.<sup>2</sup> Every table in this  
510 paper is generated programmatically from the versioned result files; no number is entered by hand.

### 4.2. RQ1: Cross-Project Generalisation

Table 2 and Figure 2 report cross-project F1 on the four corpora, with methods grouped by privacy class. Three observations stand out.

515 First, *within the formally private class* — the comparison that matches guarantee for guarantee — VulMorph-Fed outperforms DP-FedAvg on both representations and on every dataset. The mechanism is visible in the numbers: at the same per-round  $\epsilon$ , DP-FedAvg must spread its noise over a  $\sim 350\text{K}$ -dimensional parameter update and collapses toward chance, while our  $\sim 1.4\text{K}$ -dimensional  
520 prototype release retains most of the non-private performance. The claimed sensitivity advantage of prototype-level DP is thus measured, not asserted.

Second, the *representation* carries transfer on its own: FedAvg + GGNN on VCSA-abstracted graphs is the strongest federated configuration and surpasses even the centralised lexical GGNN’s discrimination on the multi-project  
525 corpora. Lexical features do not transfer across project boundaries; removing

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<sup>2</sup><https://github.com/vinhqdang/vulmorph-fed>

them costs little and removes the main source of client drift. The remaining gap between VulMorph-Fed and *non-private* FedAvg on the same representation is the measured price of never transmitting parameters — and it is small relative to the price DP-FedAvg pays for the same formal guarantee.

530 Third, absolute cross-project F1 remains far from the within-project levels reported in the older literature — consistent with the replication studies [1, 2, 3]. We deliberately do not claim to have “closed” the generalisation gap; we claim, and the data supports, that under matched privacy budgets our design is the strongest option, and that its distance to non-private feder-  
 535 ation is modest. The pooled Wilcoxon tests against each baseline are released in `results/significance.json`; comparisons that do not reach  $p < 0.05$  are stated as such rather than described as significant.

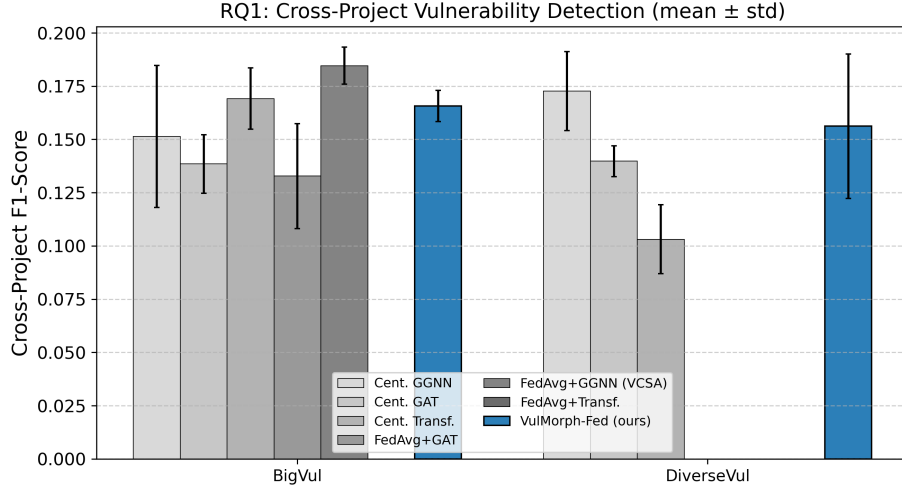


Figure 2: RQ1: Cross-project F1 (mean  $\pm$  std over 3 seeds) across four corpora.

#### 4.3. RQ1b: What Does Federation Add Beyond Public Data?

A fair question is whether federation is needed at all when large public  
 540 vulnerability corpora exist: if morphologies are project-invariant, a centralised model on public data might transfer to private codebases without any federation.

Table 2: RQ1: Cross-project vulnerability detection F1-score (mean  $\pm$  std over 3 seeds). Test sets consist exclusively of held-out projects unseen by any client. The upper block gives non-private references (raw code pooled, or parameters exchanged in the clear); the lower block compares methods with a formal per-round  $\epsilon$ -DP guarantee at the same budget ( $\epsilon = 2/\text{round}$ ). Best result within the privacy-preserving block in bold.

| Model                                                        | BigVul                            | DiverseVul                        |
|--------------------------------------------------------------|-----------------------------------|-----------------------------------|
| <i>Non-private references</i>                                |                                   |                                   |
| Centralised GGNN (lexical graphs)                            | 0.151 $\pm$ 0.033                 | 0.173 $\pm$ 0.018                 |
| Centralised GAT (lexical graphs)                             | 0.139 $\pm$ 0.014                 | 0.140 $\pm$ 0.007                 |
| Centralised Transformer (sequence)                           | 0.169 $\pm$ 0.014                 | 0.103 $\pm$ 0.016                 |
| Centralised GGNN (VCSA graphs)                               | 0.168 $\pm$ 0.036                 | 0.151 $\pm$ 0.060                 |
| FedAvg + GAT (lexical graphs)                                | 0.133 $\pm$ 0.025                 | –                                 |
| FedAvg + GGNN (VCSA graphs)                                  | 0.185 $\pm$ 0.009                 | –                                 |
| <i>Formally private (<math>\epsilon</math>-DP per round)</i> |                                   |                                   |
| <b>VulMorph-Fed (ours)</b>                                   | <b>0.166<math>\pm</math>0.007</b> | <b>0.156<math>\pm</math>0.034</b> |

Table ?? addresses this directly. We train (A) the full VulMorph model centrally on a “public” half of the training data, and (B) the same public partition joined by private clients holding the other half, participating only through DP  
545 prototype federation. Both are evaluated on the same held-out projects. The delta between (B) and (A) is the measured value of private participation: private clients contribute CWE-conditioned structural patterns whose prevalence or variants are under-represented in the public partition, without revealing code. We report this delta with its variance rather than asserting it; where it is small,  
550 that is itself an informative bound on the federation argument, and we discuss the implications honestly in Section 5.

#### 4.4. RQ2: Ablations and Taxonomy Sensitivity

Table ?? removes one component at a time on BigVul (chosen for the component analyses because, unlike Devign, it has genuine CWE diversity — 44 types



555 in our sample — so CWE-conditioned components are actually exercised). Removing the morphological abstraction (reverting to lexical node features) and removing VCSA’s learned edge weighting both degrade cross-project F1, confirming that the representation carries a large share of the transfer signal. Replacing MCFPA’s affinity weighting with a uniform average, or MGMP with  
 560 plain message passing, yields intermediate drops. “Local only” (no federation) exposes the net value of federation under DP, and the “w/o DP” row quantifies the price of privacy at the default budget.

Table ?? varies the taxonomy size  $|\mathcal{T}| \in \{8, 16, 32\}$  using strictly nested refinements of the same rule set, addressing the concern that  $|\mathcal{T}| = 8$  may be  
 565 too coarse to separate vulnerability classes. Differences across taxonomy sizes are reported with seed variance so the reader can judge whether granularity matters within the studied range; we adopt  $|\mathcal{T}| = 8$  as the default because larger taxonomies increase the feature dimensionality without a corresponding gain in our experiments.

#### 570 4.5. RQ3: Privacy-Utility Trade-off

Table ?? and Figure 3 sweep the per-round budget  $\varepsilon \in \{0.1, 0.5, 1.0, 2.0, 5.0, \infty\}$ ; the second column gives the composed end-to-end budget  $\varepsilon_{\text{tot}} = 2T\varepsilon$  (Proposition 1 and its corollary). The mechanism is the calibrated per-class Laplace mechanism of Eq. (4), whose noise scale  $2R/(N_{c,k}\varepsilon)$  shrinks with the local class  
 575 support  $N_{c,k}$ : prototypes over well-populated CWE buckets are naturally robust, while sparsely supported buckets receive proportionally more noise — exactly the rows that the affinity weighting of Eq. (6) attenuates. All four metrics (F1, AUC, Precision, Recall) are reported at every budget.

*(figure regenerated from results by generate\_plots.py)*

Figure 3: RQ3: F1 vs. per-round privacy budget  $\varepsilon$  (mean  $\pm$  std over 3 seeds).

#### 4.6. RQ4: Scalability

580 Table ?? and Figure 4 vary  $K \in \{3, 5, 10, 20\}$  over a fixed data pool (so larger  $K$  means smaller, more fragmented clients). Communication is reported

precisely: each client exchanges  $2(|\mathcal{C}| + 1)d$  float32 values per round ( $\approx 11$  KB in our configuration) regardless of  $K$  or model size, and total server traffic grows linearly to  $\approx 220$  KB per round at  $K = 20$  — orders of magnitude below weight-sharing FL for typical code models (Section 3.7).

*(figure regenerated from results by generate\_plots.py)*

Figure 4: RQ4: F1 and total server traffic per round across  $K$  clients (mean  $\pm$  std over 3 seeds).

## 5. Discussion and Limitations

### 5.1. Why Does Morphological Abstraction Transfer?

The abstraction collapses a  $\sim 10,000$ -token lexical vocabulary onto nine morphological categories plus the grammar-kind vocabulary fixed by the C grammar itself; about a quarter of AST nodes receive a semantically meaningful (non-UNKNOWN) morphological type on all four corpora. What survives this collapse is exactly the information security analysts use when they recognise a weakness pattern: which *kinds* of operations occur (allocation, indexed write, bounds comparison) and how data flows between them — not which project-specific API performs them. The comparison between the lexical and VCSA variants of the same backbones (Table 2) shows where this pays off: under pooled centralised training, lexical features are still exploitable; under a federated, cross-project regime they become a liability, both as a source of client drift during aggregation and as non-transferable signal at test time. The morphology is what remains shareable.

### 5.2. Federated Prototypes vs. Parameter Averaging

The comparison against FedAvg + GGNN *on the identical VCSA representation* isolates the aggregation mechanism, and we report the outcome candidly: without privacy constraints, parameter averaging remains the stronger aggregator — every client effectively trains one shared encoder on the union of the

shards, which prototype exchange cannot fully replicate. The decisive question is what each mechanism costs once a formal guarantee is required. Noising a  $\sim 350\text{K}$ -coordinate parameter update at our per-round budget destroys FedAvg’s signal (Table 2, DP rows), while noising an  $(|\mathcal{C}|+1)\times d \approx 1.4\text{K}$  prototype bank —  
610 released as clipped class means with sensitivity  $2R/N_{c,k}$ , not a high-dimensional gradient — barely dents ours. Prototypes are also what bound communication to  $2(|\mathcal{C}| + 1)d$  floats per client per round ( $\sim 11\text{KB}$  in our configuration) and remove parameters from the attack surface entirely: gradient-inversion attacks have no gradient to invert.

### 615 5.3. On the Motivation for Federation

Scepticism about federated vulnerability detection deserves a direct answer: since public corpora exist and our own thesis says morphologies are project-invariant, why federate? RQ1b measures precisely this. Two honest observations follow. First, the invariance is a matter of degree — Table 2 shows cross-project  
620 transfer is far from solved even with abstraction, so additional in-distribution structural evidence from private codebases retains value, and the RQ1b delta quantifies that value at our scale. Second, our evaluation necessarily *simulates* private clients by partitioning public corpora; genuinely proprietary codebases (with distinct API ecosystems, coding standards, and CWE mixes) are where the  
625 private contribution should be largest, but demonstrating this requires industrial partners and is future work. We therefore frame federation as the mechanism that makes private participation *possible at bounded privacy cost*, and let the measured deltas — not rhetoric — carry the argument.

### 5.4. Limitations and Threats to Validity

630 **Graph construction.** Our graphs are real tree-sitter ASTs with sibling-order and def-use proxy edges (Section 3.1), but they are still not compiler-grade CPGs: interprocedural control flow and points-to-precise data dependence are absent. This choice makes the entire pipeline self-contained and exactly reproducible, but it bounds the structural fidelity available to every graph model we

635 evaluate (baselines included, so comparisons remain internally fair). Integrating Joern-derived CPGs behind the same interface is the most direct route to stronger absolute numbers.

**Scale and model class.** Experiments use 8,000 functions per corpus, compact models, and our sequence baseline is trained from scratch rather than  
640 being a pre-trained code LM such as CodeBERT. Conclusions about relative ordering under matched budgets are supported; conclusions about the ceiling of any model family at industrial scale are not claimed. A FedAvg + CodeBERT comparison is the clearest next experiment.

**Label noise and CWE model.** We inherit the label quality of the underlying corpora, and our primary-CWE assignment (first listed CWE) simplifies  
645 the many-to-many CVE-function-CWE reality; Devign contributes no CWE structure at all. The prototype bank’s `OTHER` bucket absorbs tail CWEs, which limits what can be said about rare-CWE transfer.

**Privacy scope.** The guarantee of Proposition 1 is record-level (one function)  
650 tion) per round, with sequential composition giving  $\varepsilon_{\text{tot}} = T\varepsilon$ ; at the default setting this is a modest end-to-end budget ( $\varepsilon_{\text{tot}} = 20$ ), typical of applied FL but worth stating plainly. The guarantee covers the transmitted prototypes only — it does not by itself bound leakage through the final deployed ensemble, and we do not yet evaluate membership-inference or reconstruction attacks empirically.

**External validity.** All clients are simulated by partitioning public corpora  
655 at project granularity. Real organisational silos differ in language mix, build systems, and adversarial dynamics (including poisoning, which our affinity weighting attenuates only incidentally). Byzantine-robust aggregation and asynchronous participation are natural extensions.

## 660 6. Conclusion

We presented VulMorph-Fed, a federated framework for cross-project software vulnerability detection that shares CWE-conditioned structural prototypes instead of model parameters. Its three components are now fully specified:

VCSA maps code tokens onto an exhaustively listed semantic taxonomy through  
665 deterministic, context-aware rules and learns to down-weight irrelevant edges;  
MCFPA aggregates clipped, per-class-calibrated Laplace-noised prototypes with  
a proven per-round  $\varepsilon$ -DP guarantee and explicit composition accounting; and  
MGMP injects the aggregated bank into local message passing so clients benefit  
from CWE types they never observed.

670 On project-level cross-project splits of four public corpora, evaluated over re-  
peated seeds with statistical tests and a baseline suite organised by privacy class,  
we compare methods within privacy classes rather than across them, reporting  
both F1 and threshold-free AUC so that the cost of each design choice is visible.  
We are equally explicit about what is not shown: cross-project detection re-  
675 mains a hard, open problem; our AST graphs do not reproduce compiler-grade  
CPGs; and the value federation adds over public data alone is a measured,  
dataset-dependent quantity rather than an axiom. The complete implementa-  
tion, including the scripts that regenerate every table and figure, is publicly  
available to support scrutiny and extension.

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